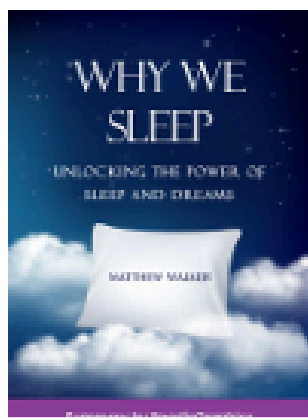




WHY WE SLEEP

Unlocking the Power of Sleep and Dreams

MATTHEW WALKER, PHD



The Big “So What”

All living creatures need sleep. The question is: *why*? In this book, neuroscientist and sleep expert Prof Matthew Walker presents groundbreaking discoveries about sleep and how it affects all aspects of our physical, mental and emotional health, including our longevity.

In this summary, we'll present the key findings in 3 parts: (i) What is sleep, (ii) Why you should sleep, and (iii) Improving your sleep.



What is Sleep?



Sleep Architecture

Every night, your brain switches between 2 types of sleep:



Non-rapid eye movement (NREM) sleep is characterized by deep, slow brainwaves that're 10x slower than when you're awake.

- The thalamus is the sensory gatekeeper that decides if sensory signals are allowed to your brain's cortex to become a conscious perception. During deep NREM sleep, this gate is closed, resulting in a sensory blackout. This allows your cortex to relax.
- Your entire brain is now fully aligned to distill and transfer selected memories from your short-term memory (in your frontal lobes) to your long-term memory (at the back of your brain). The slow, deep-sleep brainwaves are like couriers sending info packets within your brain.

KEY QUOTES

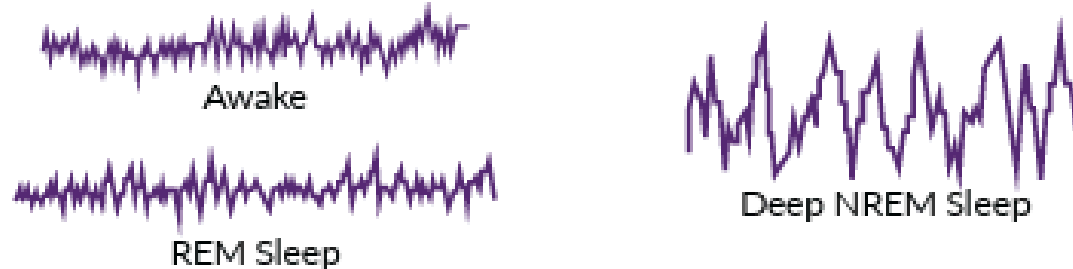
“Sleep is the single most effective thing we can do to reset our brain and body health each day—Mother Nature’s best effort yet at contra-death.”

“What you are actually experiencing during deep NREM sleep is one of the most epic displays of neural collaboration that we know of. Through an astonishing act of self-organization, many thousands of brain cells have all decided to unite and ‘sing’ or fire, in time.”

 Rapid eye movement (REM) sleep is characterized by faster brainwave activity that's similar to when you're awake. Your body is still resting, but your thalamus is now open to *internal* sensory inputs (rather than external ones). When you dream, your brain basically combines pockets of feelings, info, memories, motivations etc. into a giant movie screen, and your eyes may move rapidly as you dream. However, your voluntary muscles are deactivated (muscle atonia) to protect you from acting out your dreams unconsciously. Your sense of time seems dilated, i.e. you may feel like an hour has passed when it's only been 10 mins.

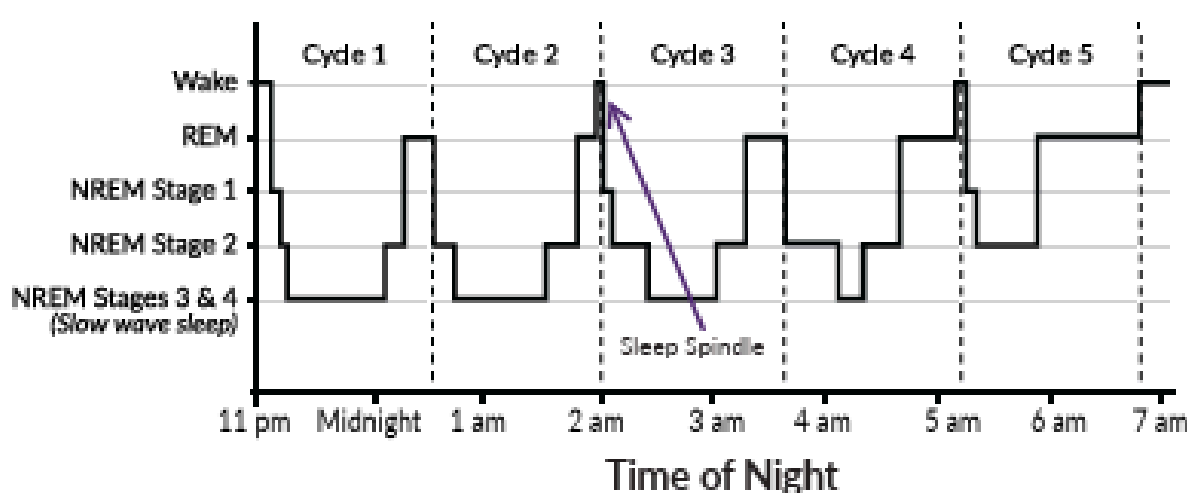
 When we're awake, we constantly receive new sensory inputs from the external environment. NREM sleep reinforces and stores those raw data and skills. REM sleep integrates these ingredients by connecting them with one another *and* with our past experiences and knowledge, to improve our mental model of how the world works, develop new insights and solve problems innovatively.

Brainwave patterns



 A typical night's sleep happens in recurring 90-minute NREM and REM cycles. When you fall asleep, you'll first enter light NREM sleep (stages 1-2), then deeper NREM sleep (stages 3-4). As the night progresses, a greater fraction of each cycle is spent in REM sleep, with the last part being richest in REM sleep.

Sleep Architecture



KEY QUOTES

- Occasionally, you'll have a sleep spindle—a burst of brain-wave activity—which usually comes at the end of a slow wave before you enter REM sleep. The spindles block off external noises, so people with more/stronger spindles sleep more soundly.
- NREM and REM serve different but equally-vital functions. When you don't get full 8 hours of sleep, you lose out on chunks of NREM or REM sleep with serious repercussions (which we'll explain later).



Understanding Your Sleep Rhythm

Your wakefulness and need for sleep are regulated by 2 key factors.



The Circadian Rhythm is your internal body clock that runs in an approximate 24-hour cycle.

- It's regulated by melatonin, which is produced by the suprachiasmatic nucleus in the brain, which calibrates your body clock using the amount of light.
- Melatonin is typically released into your bloodstream from dusk to tell your body to prepare to sleep. The amount is slowly reduced through the night, until sunlight at dawn tells your brain to stop melatonin release. Besides daylight, other reliably recurring signals (e.g. temperature change, social interactions) can also be used to reset our biological clock.
- Every human has a different internal rhythm, i.e. we feel most awake at different times of the day due to our genes. Some feel energized in the mornings (the morning larks) while others are most functional at night (the night owls). Unfortunately, our society favors daytime work—night owls may be misunderstood as lazy and end up burning the midnight oil to catch up on their unproductive workday.



Adenosine is a chemical that makes you sleepy.

- It builds up progressively when you're awake; most people will feel "sleep pressure" after being awake for 12-18 hours. It decreases as you sleep and is usually purged fully after 8 hours of sleep.

KEY QUOTES

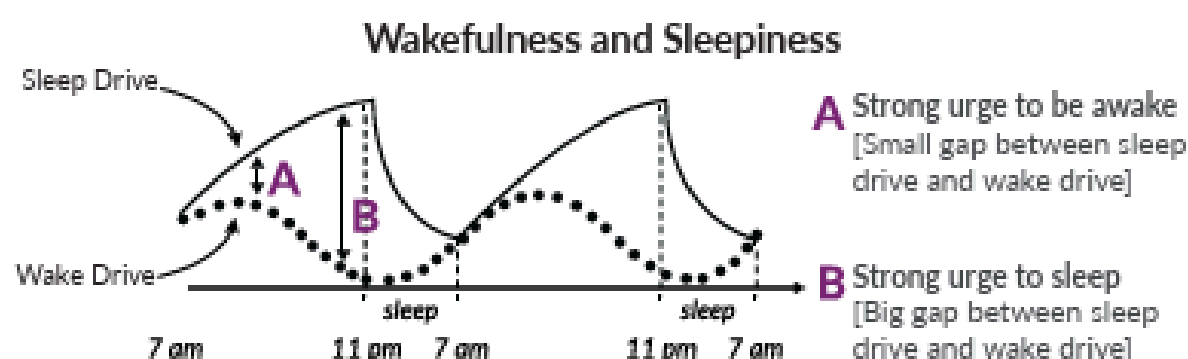
"When it comes to sleep, there is no such thing as burning the candle at both ends—or even at one end—and getting away with it."

"Wakefulness and sleep are...under the control of the circadian rhythm, and not the other way around."

"We require more supple work schedules that better adapt to all chronotypes, and not just one in its extreme."

- Caffeine can dull the effects of Adenosine, but doesn't stop the chemical from building up in your body. So, you may feel worse once the effect of caffeine wears off. If you then consume more caffeine to counter the fatigue, a dependency cycle is created. Caffeine taken in the evening can disrupt sleep since 50% of caffeine will still be in your system 5-7 hours later. Caffeine is found in coffee, teas, energy drinks, dark chocolate and certain drugs. Even decaffeinated coffee usually contains 15-30% of caffeine in regular coffee.

+ The 2 mechanisms above operate independently, with the circadian cycle powering your “wake drive” and adenosine powering your “sleep drive”. The combined effects mean that after a proper night's rest, you'll feel wide awake at noon since there's only a small gap between the 2 drives. At 11pm, you'll feel an urge to sleep since there's now a huge gap between the 2 drives.



🧐 You're probably not getting enough quality sleep if: (i) you can fall asleep a few hours after waking up (e.g. at 10-11am), (ii) you can't function optimally without caffeine before noon, (iii) you can't wake up naturally at the time you set your alarm, and/or (iv) you keep rereading something because you can't focus.

Who Sleeps and How?

🧐 Every animal species sleeps, including worms, insects, fish, reptiles, amphibians, mammals and humans.

- All species have NREM sleep, but many don't seem to have REM sleep (e.g. insects, fish, amphibians and most reptiles). Only birds and mammals—which are newer on the evolutionary timeline—have full-blown REM sleep. Compared to other animals, humans need much less sleep, and we have a disproportionately high amount of REM sleep. Walker theorizes that humans have evolved with fewer hours of more efficient/intensive sleep.

KEY QUOTES

“Without exception, every animal species studied to date sleeps, or engages in something remarkably like it...Sleep is universal.”

“Regardless of the amount of recovery opportunity, the brain never comes close to getting back all the sleep it has lost.”

- **There's no way to fully recover from lost-sleep.** When you're recovering from sleep-deprivation, your body tends to prioritize NREM over REM recovery. However, once you've recovered on your NREM, the proportion of REM sleep will increase.

 Today, most humans are not sleeping adequately nor optimally. In most modern societies, people usually sleep in a monophasic pattern, with 1 stretch of only <7hrs a night. **Generally, an adult human needs 8 hours of sleep.** In societies without electricity (e.g. hunter-gatherer tribes), people sleep in a biphasic pattern, with a stretch of 7-8 hrs at night (2-3 hrs after sunset till dawn) *plus* a 30-60 minute mid-afternoon nap (to address the natural dip in alertness). Some modern cultures still observe siestas, and people who kept Greek siestas were found to have a 37% lower risk of death from heart disease compared to those who didn't.

 Our sleep hours and patterns change across our lifespan:

- Babies. Mammals' brain construction is done most heavily during REM sleep, which is why infants sleep so much. Newborn rats that were denied REM sleep suffered retarded gestational progress. Since alcohol is a strong suppressor of REM sleep, infants whose mothers drink heavily have much lower levels of vibrant brain activity compared to those whose mothers don't drink.
- Infants and young children have polyphasic sleep, i.e. they sleep for many short periods throughout the day. As their suprachiasmatic nuclei develop and gain a rhythm, they shift to a daily rhythm, then a biphasic pattern and a finally a monophasic sleep pattern by late childhood. As a child grows up, total sleep time drops and the proportion of REM-to-NREM sleep decreases.
- During late childhood and adolescence, cognitive and critical thinking skills are honed during deep NREM sleep to affect maturity and rationality. Teens have a later circadian cycle, preferring to stay up later and wake later. Asking them to sleep at 10pm is like asking adults to sleep at 7-8pm. Unfortunately, sleeping late and waking early for school mean that our teens are sleeping much less than they would naturally.

KEY QUOTES

"Humans are not sleeping the way nature intended. The number of sleep bouts, the duration of sleep, and when sleep occurs have all been comprehensively distorted by modernity."

"REM sleep is not optional during early human life, but obligatory."

"An infant brain without sleep will be a brain ever underconstructed."

"Deep sleep may be a driving force of brain maturation, not the other way around."

- Mid-life and old-age. It's not true that the elderly need less sleep. Most seniors are actually sleep-deprived due to 3 factors:
 - (i) They have less *quantity/quality* of deep NREM sleep. NREM brainwaves start to become smaller and fewer from our late 20s. By 70 years old, we'd have lost 80-90% of the deep REM sleep we enjoyed as a teen.
 - (ii) They *sleep less efficiently* since they wake up more frequently at night (due to illnesses, a weaker bladder etc.).
 - (iii) They develop an *earlier circadian rhythm*. Instead of sleeping early, they may take an evening nap which disrupts their night sleep, then wake up pre-dawn due to their cycle. Seniors can try using Melatonin to sleep better, and push their circadian rhythm back by getting bright-light exposure in the late afternoon instead of mornings.



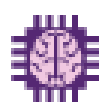
Why You Should Sleep

Sleep is the ultimate therapy and enhancer to provide a range of benefits that medicine and other interventions cannot.



Sleep Benefits the Brain

Sleep is essential to our brain and delivers 3 main cognitive benefits.



Sleep improves memory. The hippocampus in our brain stores short-term memory and has a limited capacity, while the cortex stores long-term memory with a larger capacity. In NREM sleep:

- Short-term memory is transferred to the cortex, helping to (i) free up space so we can absorb new info and (ii) save useful info for the long term. In fact, after sleeping, you may recall something you previously forgot.
- You discard useless memories to make space for those worth remembering. You can use *targeted memory reactivation* to lock in vital info: before going to bed, review what you've

KEY QUOTES

"That older adults simply need less sleep is a myth."

"There is much that sleep can do that we in medicine currently cannot."

"Sleep has proven itself time and again as a memory aid: both before learning, to prepare your brain for initially making new memories, and after learning, to cement those memories and prevent forgetting."

learned and mentally “tag” those you’d like to remember or improve, then leave the rest to your brain.



Sleep improves motor task proficiency. Once you learn to ride a bike, you won’t forget it—that’s “muscle memory” at work, though the memory resides in your brain (not your muscles). You may struggle with a motor task (e.g. playing a difficult piano sequence), but can suddenly do it flawlessly after sleeping. That’s because during sleep, your brain continues to practice the motor skills and work out the kinks. Sleep spindles are associated with better memory effects (including motor memories) and are most concentrated in the last cycle of deep NREM sleep. This means that when you wake up early, you’re actually losing valuable sleep hours.



Sleep improves creativity. REM sleep connects your different memories, experiences and skills to create new ideas and insights.



Sleep-deprivation Harms the Brain

A lack of sleep causes brain impairment, with repercussions for all aspects of your mental and physical health.



Sleep-deprivation worsens focus/concentration.

- After being awake for 19hrs, you’re as cognitively impaired as someone who’s legally drunk. After 10 days with 7hrs of sleep, your brain is as dysfunctional as someone who’s been awake for 24hrs. Only a rare 1% of the population is genetically-enabled to survive on 6hrs of sleep with minimal impairment. Power naps can help defer fatigue if sleep-deprivation hasn’t sunk in, but they can’t replace regular sleep.
- When you’re tired, you may also fall into microsleep, i.e. the brain shuts down momentarily and you lose all awareness for 1-2 sec. This can be fatal if you’re engaged in risky activities like driving. In the US, the number of traffic accidents from fatigue-related errors are higher than those from alcohol and drugs combined.
- Even if you sleep for 3 full nights, you can’t recover from a week of sleep-deprivation. Yet, we’re unaware of the effects

KEY QUOTES

“Your brain will continue to improve skill memories in the absence of any further practice.”

“Sleep provides a nighttime theater in which your brain tests out and builds connections between vast stores of information.”

“No facet of the human body is spared the crippling, noxious harm of sleep loss.”

“The human mind cannot accurately sense how sleep-deprived it is when sleep-deprived.”

of sleep-debt since we get used our worsening performance levels.



Sleep-deprivation worsens emotional control. When we lack sleep, the amygdala (our brain's emotional center) becomes more active and overpowers the prefrontal cortex (the brain's logical center). We swing between extreme negative moods (linked to aggression and bullying) and extreme positive moods (linked to risk-taking and addiction).



Sleep-deprivation impairs memory since (i) the hippocampus seems to shut down without enough sleep, and (ii) it can cause lasting damage to our DNA and learning-related genes.



Sleep-deprivation plays a role in mental illnesses. By worsening mood swings, it can reinforce *psychiatric conditions* (and vice versa) in a vicious cycle. Sleep-loss is also correlated with *Alzheimer's Disease*. During NREM sleep, the glymphatic system flushes out toxins from the brain (including those responsible for Alzheimer's). A lack of sleep could create a buildup of toxins, which in turn disrupts NREM sleep in a vicious cycle.



Sleep-deprivation Harms the Body

Sleep-deprivation damages every aspect of our physiology and is linked to a host of diseases (e.g. cancer, heart diseases) and a shorter lifespan.



Cardiovascular System. Insufficient sleep is widely linked to an *overactive sympathetic nervous system*, i.e. the body is in a stressful fight-or-flight state. This leads to (i) increased heart rates, blood pressure, levels of cortisol (a stress hormone) and atherosclerosis, and (ii) decreased HDL (good cholesterol) and growth hormone (to repair blood vessels). The opposite happens during deep NREM sleep—the body relaxes while heart rate and blood pressure falls. A lack of sleep is associated with a 45% increase in the risk of developing and/or dying from coronary heart disease in the next 25 years. Just sleeping 1-2 hours less for 1 night can increase heart rate and systolic blood pressure.



Insufficient sleep leads to **weight gain**, since (i) you're likely to eat more and (ii) your body can't handle the calories as

KEY QUOTES

"The under-slept brain swings excessively to both extremes of emotional valence, positive and negative."

"Sleep is not like the bank. You cannot accumulate a debt and hope to pay it off at a later point in time."

"Sleep is...the foundation on which the other two health bastions [diet and exercise] sit. Take away the bedrock of sleep, or weaken it just a little, and careful eating or physical exercise become less than effective."

"Unhealthy sleep, unhealthy heart. Simple and true."

effectively. Chronic sleep deprivation is a key contributor to type 2 diabetes.

- When you lack sleep, you eat excessively due to an increase in chemicals that signal hunger and carbohydrates-craving, and a decrease in chemicals that signal that you're full.
- You also feel lethargic and are less likely to exercise. Even if you lose weight, sleep-deprived people tend to lose lean body mass (vs those with enough sleep tend to lose more fat).

 **Reproductive System.** Sleep-deprived males have 29% lower sperm count, smaller testicles, and decreased testosterone (which in turn affect libido, bone density and muscle mass). Sleep-deprived females have 20% lower follicular-releasing hormone (required for conception), more infertility issues, and higher risks of miscarriages. After just 1 night of sleep shortage, participants' face in a study were rated less attractive.

 There's a clear correlation between sleep and immunity.

- Sleep is vital in the prevention and fight against cancer. When we sleep, our immunity system releases "killer cells" to remove bad foreign elements in our bodies. After just 1 night of sleep loss, you can lose 70% of your killer cells. Regularly sleeping ≤ 6 hours a night increases your risk of developing cancer by 40% compared to those who sleep ≥ 7 hours.
- Sleep-deprivation also increases the risks of cancer inflammation. Lab rats were injected with malignant cells and split into 2 groups (with normal vs insufficient sleep). The sleep-deprived rats had (i) faster and bigger cancer growth, and (ii) more aggressive cancer cells that metastasized or spread to other organs.

 **Genes and DNA.** Every cell in our body contains DNA that defines who we are and tells our body what to do. Reducing sleep for just 1 night can reduce the genetic activity for healthy men and women. Chronic sleep-loss can even change our genetic structure. Telomeres are found at the end of your DNA chromosomes (like the tip of your shoelaces). The more sleep-deprived you are, the more damaged your telomeres become to increase the risks of changing your genetic makeup.

KEY QUOTES

"Inadequate sleep is the perfect recipe for obesity: greater calorie intake, lower calorie expenditure."

"Chronic sleep loss will erode the very essence of biological life itself: your genetic code and the structures that encapsulate it."



Benefits of Dreaming

Neuroscience has shed light on how, what and why we dream. Using MRI scans, scientists can now predict what you're dreaming. **Dreams offer real benefits for our survival and well-being.**



Dreams reduce the pain from traumatic events. Dreams capture 35-55% of the emotional concerns and themes from our waking lives. Deep REM sleep is the *only* time when the stress-related chemical noradrenaline is totally absent from your brain. This allows you to process painful memories without being crippled by the visceral feelings. However, this *only* works if you dream about the emotional themes associated with the trauma; REM sleep or dreams about unrelated themes won't help you to move past your trauma.



Dreams help us to decode facial expressions, which are one of the most important external cues to help us discern friend from foe, and help us to interact effectively with others. People with a full night's sleep were able to accurately decipher others' emotions based on their facial expressions. The same people, when deprived of sleep, were less able to do so and tended to see others as hostile or aggressive.



Dreams improve problem-solving and creativity. During deep NREM sleep, brain activity drops. But once we enter REM sleep and start dreaming, the visual, motor, memory and emotional centers of our brain are activated while the prefrontal cortex and logical centers are switched off.

- This means that in your dreams, your brain fuses past and present knowledge and experiences in creative/abstract ways, without the limits of logical filters. Your dreaming brain can apply an idea to your entire storage bank and make novel connections that your waking brain can't conceive.
- For example, after years of seeking a way to organize the world's elements, Dmitri Mendeleev literally dreamt up the solution in 1869. In his dream, he saw how all the elements fell neatly into a table. His dreaming mind could integrate his years of research into the period table of elements which we now use—something his waking mind couldn't.

KEY QUOTES

"It was not...time...that healed all wounds, but instead it was time spent in dream sleep that was providing emotional convalescence."

"REM-sleep dreaming is informational alchemy."

"Our human memories are...richly interconnected in webs of associations that lead to flexible, predictive powers."



Sleep Disorders and Death

Sleep disorders can disrupt our lifestyle and well-being.



Somnambulism refers to sleep disorders that involve movement, including sleepwalking, sleep talking etc. This happens during NREM sleep (not REM sleep), when spikes in electric pulses in the nervous system cause the person to be stuck between sleep and wakefulness. Sleepwalkers often perform familiar routines unconsciously (e.g. brushing their teeth, drinking water).



Insomnia. Unlike sleep deprivation (where the person is capable of sleep but doesn't get enough opportunity to sleep), insomniacs set aside enough time (7-9 hours) for sleep, but have difficulty falling asleep and/or staying asleep for more than 3 months. The most common causes of insomnia are emotional concerns or distress, which are linked to an overactive sympathetic nervous system that keeps the brain overly-active. Even when asleep, insomniacs have shallower deep NREM brainwaves and more fragmented REM sleep.



Narcolepsy comes with 3 key symptoms: (i) extreme sleepiness in the day (e.g. irresistible urges to sleep), (ii) sleep paralysis (i.e. loss of the ability to move/talk when you wake up), and (iii) cataplexy (sudden loss of muscle control e.g. face sagging or limb muscles going limp). Narcoleptic patients are stuck between sleep and wakefulness because they lose the bulk of the brain cells that produce orexin, the chemical that signals wakefulness.



At the extreme, sleep deprivation can cause death. In experiments, sleep-deprived rats died after 15 days without any sleep/REM sleep, or after 45 days without NREM sleep. Notably, the sleep-deprived rats (i) lost body mass, (ii) couldn't regulate their body temperature, and (iii) had damaged immune systems resulting in skin infections and internal damage (e.g. hemorrhaging, stomach ulcers). Death from sleep deprivation is uncommon in humans, though patients with fatal familial insomnia (a rare inherited condition) degenerate and die within 10 months after they lose all ability to sleep.

KEY QUOTES

"Simply put, the insomnia patients could not disengage from a pattern of altering, worrisome, ruminative brain activity."

"The shorter your sleep, the shorter your life span...The elastic band of sleep deprivation can stretch only so far before it snaps."



Improving Sleep Quality & Quantity

On average, half of all adults in developed countries aren't getting the recommended hours of sleep. We can systematically improve our sleep quality and quantity at individual, educational, organizational, public and societal levels.



Individual Lifestyle and Sleep Habits

We can address key barriers to sleep and improve our sleep hygiene.



Light. Modern electric light disrupts our circadian rhythm by delaying the release of melatonin by an average of 2-3 hours in the evenings. Any light can disrupt your circadian rhythm, but blue LED lights—including those from computer and smartphone screens—have the worst impact. To improve sleep quantity/quality:

- Dim the lights in the rooms where you spend your evenings
- Use blackout curtains to keep your bedroom completely dark
- Use software to decrease blue LED lights from electronic devices



Besides natural light, the body uses the dip in nighttime temperature as a signal to prepare for bed. However, air-con, heaters and blankets mask such natural temperature fluctuations. Generally, it's easier to fall sleep in a cold room than a warm one—the ideal temperature for sleep is around 65°F or 18.3°C. You can reduce your core body temperature by ejecting heat via your hands, feet and head, thus facilitate sleep:

- Take a hot bath or warm your hands/feet, to draw heat to the skin surface and reduce core body temperature;
- Splash water on your face using your hands; and/or
- Stick your hands/feet outside the blankets or wear a cooling body suit.

KEY QUOTES

"What, precisely, about modernity has so perverted our otherwise instinctual sleep patterns, eroded our freedom to sleep, and thwarted our ability to do so soundly across the night?"



Caffeine. Earlier, we've explained the effects of caffeine. Avoid consuming caffeine from late afternoon.



Alcohol does *not* help you to sleep better. It's a sedative that blocks the brain's neurons from firing properly, thus suppressing REM sleep and causing us to wake up throughout the night. Alcohol disrupts sleep and harms our memory and learning abilities. The body takes several hours to breakdown and discharge alcohol, so it's best to abstain from alcohol.



In modern societies, most people force themselves to wake up using alarms. This actually creates stressful spikes in heart rates, cortisol and blood pressure levels. Ideally, you should wake up naturally at the same time daily, without alarms. If you must use an alarm, then wake up once it goes off; don't put yourself through a series of stresses by using the snooze function.



Regular exercise helps to improve the quality/quantity of sleep, which in turn improves your fitness and energy levels. However, avoid exercising 2-3 hours before bed as it increases core body temperature to make it harder to fall asleep.



Eat a normal diet and don't go to bed feeling too full/hungry. Sleep is generally disrupted if you consume too few calories (<800) or too much carbohydrates (>70% of all energy intake).



Avoid sleeping pills since they're sedatives that affect your brain in the same way as alcohol. Such drug-induced sleep lacks the deep brainwaves found in healthy sleep, resulting in next-day grogginess.

- In fact, people who use prescription sleep medications are much more likely to die and to develop cancer than those who don't. They're also linked with higher rates of stroke/heart disease, infections and accidents.
- Until scientists discover a drug that can induce natural sleep without harmful side effects, Walker recommends sleep therapy such as the Cognitive Behavioral Therapy. These involve good lifestyle habits and sleep hygiene, such as (i) adopting a consistent daily sleep cycle, (ii) reducing bedroom temperature, noise/light levels, (iii) avoiding alcohol, caffeine, naps and exercise just before sleep, and (iv) avoiding heavy mental activities or anxiety-provoking thoughts before bed.

KEY QUOTES

"Alcohol sedates you out of wakefulness, but it does not induce natural sleep."

"Sleep and physical exertion have a bidirectional relationship."

"No past or current sleeping medications on the legal (or illegal) market induce natural sleep."



Ideas for the future. We can use technology to detect/prevent sleep-deprived driving, design intelligent household lights that match our circadian rhythms, or use body trackers to identify how sleep correlates with our activity, heart rate, mood, etc.



Addressing Sleep at Systemic Levels



The Workplace. Longer work hours don't mean productivity. Sleep-deprived employees show lower levels of creativity, motivation, efficiency and effectiveness. They're also less sociable, emotionally stable and ethical/honest. Sleep-deprived leaders are less engaging and inspiring, and create a negative rippling effect on their staff.

- Inadequate sleep can cost companies \$2,000-\$3,500 per employee or tens of millions in annual losses. By allowing flexible work hours to fit employees' natural body clocks, encouraging naps and proper rest, companies can deliver better results.
- Many medical institutions still require doctors and nurses to work long hours at a stretch, especially during their years in resident training. Fatigue has contributed to an alarming number of serious/fatal medical errors and accidents. These can be reduced simply by limiting each shift to 16 hours, with at least 8-hour rest opportunities in between. There's simply no reason to continue with the existing practices of sleep-deprivation which hurt both patients and medical staff. Sleep hygiene can also be used to help patients, e.g. change harsh white lights to warm yellow lights, provide earplugs/eye-masks etc.



School Schedules. To match parents' schedules, schools often start at 8am or earlier. Kids may have to wake up at 5:30am to catch the school bus which, given their circadian rhythms, is akin to an adult waking up at 3:30am. We're robbing our kids of the vital hours of REM-rich late-morning sleep that's critical for their learning, development and well-being. Chronic sleep deprivation is also harmful to adolescents who're the most susceptible to mental illnesses. Schools that start later have seen improvements in attendance, students' grades, IQ, emotional stability and even life expectancy (due to fewer accidents). It's time to review our existing school schedules.

KEY QUOTES

"Under-slept employees are not only less productive, less motivated, less creative, less happy, and lazier, but they are also more unethical."

"Ounces of sleep offer pounds of business in return."

"There's simply no evidence-based argument for persisting with the current sleep-anemic model of medical training, one that cripples the learning, health, and safety of young doctors and patients alike."

"It is clear that a tired, underslept brain is little more than a leaky memory sieve, in no state to receive, absorb, or efficiently retain an education."

"We as a society must work toward dismantling our negative and counterproductive attitude toward sleep."



Education and public/societal interventions. Sleep is no less critical than diet and exercise, and should be made a mandatory topic in education, to build awareness and encourage better sleep practices. At a public/societal level, we should also consider public campaigns to improve sleep awareness, enact laws against drowsy driving, introduce health insurance credits for people who with consistent/quality sleep hours, etc.



Other Details in the Book to Look out For

This book is packed with useful background, theories, examples and discussions on sleep and its implications, including:

- Evolutionary and scientific information about sleep;
- How NREM/REM sleep relate to various biological functions and lifestyle factors; and
- Detailed insights, tips and ideas on how we can improve our quality and quantity of sleep.

For more details, please visit www.sleepdiplomat.com.

About the Author



Dr. Matthew Walker is an English scientist and professor of neuroscience and psychology at the University of California. His research focuses on the impact of sleep on human health and disease. Previously, he was a professor of psychiatry at Harvard Medical School. He is also the founder and director of the Center for Human Sleep Science.

He has received numerous funding awards from the National Science Foundation and the National Institutes of Health, and is a Kavli Fellow of the National Academy of Sciences. He has published more than 100 scientific research studies and has been featured on numerous television and radio programs.

KEY QUOTES